

Investigating Class Imbalance and Feature Importance for Predictive Diabetes Analysis

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Abstract—Predictive modeling plays a critical role in the comprehension and diagnosis of prevalent chronic medical conditions, such as Diabetes. This empirical case study evaluates the impact of data preprocessing techniques and machine learners on diabetes classification performance. We explore the effectiveness of the XGBoost, LightGBM, Random Forest, and Logistic Regression machine learners, comparing their performances across multiple imbalanced data and categorical features' transformations. The case study results indicate Logistic Regression performs subpar compared to the XGBoost, LightGBM, and Random Forest models in all class ratio scenarios with class imbalance. LightGBM, at the 75:25 class ratio and with one-hot encoding, emerged as the best-performing model, achieving an F1-score of 0.8159, AUROC of 0.9691, and AUPRC of 0.8954. The boosting-based machine learners maintained a relatively consistent performance across class imbalances, while Logistic Regression exhibited greater sensitivity to class imbalance variations. The results of our Feature Importance analysis identify HBA1C, Blood Glucose, BMI, Age, and Smoking History as critical features of diabetes risk. Its notable, the latter feature is a social and behavioral attribute. In this preliminary study to determine an effective categorical data transformation, the Weights of Evidence (WOE) approach and one-hot encoding technique yielded comparable results. However, WOE demonstrates potential advantages in scenarios with high-cardinality categorical features. Our case study highlights the relative effectiveness of the gradient boosting and tree-based ensemble models for data-driven diabetes prediction, with a realistic constraint presence of class imbalanced data -- a common observation in medical-domain datasets. In addition, we determine that both medical testing-based features, as well as social and behavioral attributes were important features in the predictive modeling and analysis of chronic diabetes. We continue this work with big healthcare data reflecting different medical illnesses and conditions, including diabetes.

Keywords—*Feature Importance, Class Imbalance, Diabetes ADASYN, Machine Learning.*

I. INTRODUCTION

Diabetes Mellitus is one of the most prevalent chronic diseases worldwide, placing a heavy burden on modern healthcare systems [1]. The disease is known to cause several serious health complications, including kidney failure, cardiovascular diseases, and infections. The most common type of diabetes, Type II, is particularly associated with patient characteristics such as BMI, age, and physical inactivity. With the increasing availability of clinical data in the form of electronic health records (EHRs), patient reports, and disease registries, there is growing potential to improve

diabetes prediction and detection using advanced data driven approaches such as machine learning [2].

Traditional methods for diabetes prediction and detection rely heavily on statistical analysis, but modern computational advancements have introduced alternatives like machine learning. Machine learning techniques can process large clinical datasets, uncovering patterns and trends that may be too complex for humans to identify through traditional statistical approaches. With effective software integration, clinical data could be analyzed in real time as it is recorded in electronic health systems. In addition to long-term data mining capabilities, machine learning algorithms offer potential for the real-time diagnosis and risk assessment of diabetes, optimizing healthcare workflows and improving disease monitoring [3].

While machine learning holds potential for clinical data analysis and diabetes diagnosis, several challenges must be addressed. Given that diabetes affects only about 10% of the population, clinical datasets focused on diabetes often exhibit significant class imbalances. Training models on such imbalanced data can lead to biases, where the model overfits to the majority class, reducing its reliability and decision-making accuracy [4]. Additionally, categorical features in clinical datasets must be transformed into numeric formats to be processable. Simple encoding methods, such as label encoding, can introduce false ordinal relationships misrepresenting the data and further impacting model effectiveness [5].

This study examines various machine learning approaches to diabetes prediction, with a focus on class imbalance, categorical data preprocessing, and feature importance. We evaluate multiple class balance ratios to assess their impact on classification performance. Additionally, we compare two categorical feature encoding methods Weights of Evidence (WOE) and one-hot encoding to determine their influence on model effectiveness. Feature importance is explored using SHapley Additive exPlanations (SHAP) to rank patient attributes by their contribution to diabetes predictions. Our analysis also examines the performance of XGBoost, Random Forest, Logistic Regression, and LightGBM across several metrics. Beyond individual preprocessing techniques and model choices, we investigate how these factors interact to influence overall predictive accuracy.

The remainder of the paper is organized as follows: Section II reviews related studies; Section III presents the empirical methodology; Section IV describes the selected machine learners; Section V presents results and analysis; and Section VI provides conclusions and future directions.

II. RELATED WORKS

Previous studies have explored various preprocessing techniques to improve diabetes prediction with machine learning. This section summarizes related approaches and key findings from these works.

Ramadhan [6] compared two oversampling techniques, SMOTE and ADASYN, for diabetes prediction using a Support Vector Machine (SVM) model trained on a balanced version of the Karya Medika dataset. The study found that ADASYN oversampled datasets yielded a 2–4% higher accuracy than those balanced with SMOTE. Expanding on this, Rahmawati et al. [7] conducted a broader comparative analysis using approaches with SMOTE, ADASYN, and no oversampling on an imbalanced dataset across three classifiers: Logistic Regression, Naïve Bayes, and SVM. Their findings reinforced that SMOTE and ADASYN effectively improved performance, though SMOTE demonstrated slightly better metrics with 0.5–1% higher accuracies across variations.

Ramezankhani et al. [8] further explored the effects of oversampling techniques, but at diverse levels. Increments of 100% starting at the base number of minority samples and going all the way up to 700% were used to investigate what the optimal oversampling percentage was for the minority class. The study evaluated Probabilistic Neural Network, Naïve Bayes, and Decision Tree machine learners and concluded that the highest level of oversampling (700%) yielded the best F1-scores. However, they noted that accuracy declined as the F1-scores and class balance increased.

Other studies have investigated the role of multiple preprocessing steps in improving prediction accuracy. Jian et al. [9] examined machine learning models for predicting diabetes complications, incorporating several preprocessing techniques. These included missing value imputation with both removal and mean-filling approaches, random undersampling, cluster centroids, SMOTE, rescaling, and one-hot encoding. Additionally, they applied feature selection using Logistic Regression and SVM's built in methods. Their results showed that models trained with their multi-step preprocessing approach exhibited a 10% higher accuracy compared to baseline models.

Similarly, Olisah et al. [10] proposed a preprocessing framework incorporating Spearman correlation and polynomial regression for feature selection and missing value imputation. They developed a deep neural network framework specifically for their diabetes classification tasks and compared its performance with Random Forest and SVM models using the PIMA Indian and LMCH diabetes datasets. Their results demonstrated impressive accuracy, achieving 97% with Random Forest and SVM and 99% with their custom deep neural network, attributing this success to their unique preprocessing approach.

Effective data preprocessing methods have shown potential to improve the performance of diabetes prediction. For categorical feature encoding, techniques like label or one-hot encoding are widely used. However, alternative methods may increase performance. Our study builds upon existing work by incorporating Weights of Evidence (WOE), a categorical transformation technique commonly used in the finance sector but rarely applied to clinical datasets. To our knowledge, this is the first study to evaluate WOE in diabetes

analytics. Additionally, we introduce an expanded range of class balance ratios, using random undersampling and ADASYN oversampling to further investigate their effects on model performance. By comparing tree-based classifiers such as XGBoost, LightGBM, and Random Forest alongside Logistic Regression, we aim to provide deeper insights into how preprocessing techniques interact with various machine learning models in diabetes prediction.

III. METHODOLOGY

A. Case Study Data Preprocessing

For our study, we use the [11] Diabetes Clinical Dataset, originally comprising 100,000 records and 17 features. An instance is represented by a patient's reported characteristics and diabetes status (yes or no). The features consist of medical test results, as well as social and behavioral attributes.

Our preprocessing steps aim to refine the dataset to optimize its applicability for predicting diabetes. To focus on biomedical attributes relevant to diabetes status, we remove non-essential columns, such as year and location. Additionally, the clinical notes column is excluded due to its textual format.

Analyzing the age distribution among diabetic patients, we note that only 1.56% of individuals under 40 have diabetes, compared to 14.29% of those over 40. Given this imbalance, we exclude individuals younger than 40 to concentrate on predictive modeling within the higher-prevalence age group. To enhance interpretability, we consolidate the originally one-hot encoded age features into a single column with bucketed age ranges: 40-50, 50-60, 60-70, and 70-80. These preprocessing modifications result in a dataset containing 54,513 records and 10 features.

To make the dataset processable for machine learning, categorical features are transformed into numerical representations. Since label encoding can introduce artificial ordinal relationships, one-hot encoding is often preferred. However, one-hot encoding can still lead to sparse dimensionality, so we also apply Weights of Evidence (WOE) as an alternative approach. Our results and analysis section will compare the performance of both methods.

WOE quantifies the predictive relationship between categorical features and a binary target variable. It is computed as the natural logarithm of the ratio between non-events and events within each category [12]. Positive WOE values suggest a category is associated with a higher prevalence of non-events, while negative values indicate a stronger presence of events. We apply WOE to four categorical variables: gender, smoking history, race, and age.

Expanding on WOE, Information Value (IV) [13] provides a complementary feature elimination metric that assesses how strongly categorical variables predict the target outcome. IV scores range from 0.02–0.1 (weak predictors), 0.1–0.3 (medium predictors), and above 0.3 (strong predictors). Applying IV to our WOE transformed categorical features yields the following scores: 0.21 for age bucket, 0.09 for smoking history, 0.2 for gender, and 0.0009 for race. Based on these results, we exclude gender and race in our IV based model runs and evaluate model performance both with and without feature removal.

TABLE I. PREDICTIVE FEATURES

1	Gender
2	Hypertension
3	Heart Disease
4	Smoking History
5	BMI
6	HBA1C
7	Blood Glucose
8	Race
9	Age

B. Class Imbalance

To prepare data for training, we split the dataset into 80% training and 20% testing. Given the natural class imbalance, we apply a hybrid resampling approach combining random undersampling and ADASYN oversampling. Resampling is performed only on the training dataset to preserve real data instances during evaluation.

Random undersampling [14] mitigates imbalance by selectively removing samples from the majority class. ADASYN [15] is an adaptive oversampling method that generates synthetic data points for the minority class, giving it better representation. It assigns weights to instances based on their local distribution, focusing on hard to learn examples due to their sparse presence. By generating additional synthetic samples in these rare regions, ADASYN enhances the model's ability to capture complex patterns. Unlike traditional oversampling, ADASYN prioritizes the most underrepresented instances, improving classification performance in challenging scenarios.

To explore different class balance ratios, we methodologically adjusted the distribution of samples while maintaining a constant total number of training records. These variations create proportional intervals ranging from the dataset's natural imbalance to a fully balanced distribution, allowing us to assess the impact of class proportion on model performance.

TABLE II. CLASS RATIOS (ROUNDED)

N:P Ratio	Negative (N)	Positive (P)
86:14	37,373	6,237
80:20	35,016	8,594
75:25	32,708	10,902
63:37	27,256	16,354
50:50	21,805	21,805

After resampling, we apply five-fold cross-validation [16] on the training data. We chose five folds to offer a balance between reliability and computational efficiency.

C. Feature Importance

To deepen our analysis of diabetes and the significance of predictive features, we employ SHapley Additive exPlanations (SHAP) [17] and the built-in feature importance of our top performing models to rank features based on their influence on model decision-making. To ensure unbiased insights, feature importance is assessed solely on the test

dataset, preventing any artificial relationships stemming from oversampling techniques. We will apply feature importance to the two top performing models, and categorical data transformation scenarios.

D. Performance Metrics

We evaluate model performance using multiple metrics [18] to assess their ability to distinguish between diabetes and non-diabetes cases. Our primary focus is on test metrics, which provide insight into model effectiveness on real-world data. Additionally, we analyze training metrics to detect signs of overfitting. Models are assessed using the following performance measures: accuracy (Acc), precision (Prec), recall (Rec), F1-score (F1), geometric mean (G-Mean), area under the receiver operating characteristic curve (AUROC), and area under the precision-recall curve (AUPRC). These metrics collectively capture various aspects of classification performance, providing a well-rounded evaluation.

IV. MACHINE LEARNING MODELS

XGBoost (eXtreme Gradient Boosting) [19] is a gradient boosting algorithm designed for speed and accuracy. It constructs multiple decision trees while incorporating L1 and L2 regularization and parallel processing for efficient computation. To improve learning, XGBoost assigns weights to misclassified instances, allowing it to refine decision-making over iterations. Due to its regularization techniques and adaptive learning, XGBoost may be suited for imbalanced datasets.

LightGBM (Light Gradient Boosting Machine) [20] is a fast, efficient gradient boosting algorithm that differs from XGBoost by having a leaf-wise tree growth strategy rather than level-wise splitting. By splitting the leaf that reduces loss the most, LightGBM creates deeper trees, allowing it to capture complex patterns in the data. Given its fast-training speed and deep trees, LightGBM was included because of its balance between efficiency and predictive power.

Random Forest [21] is an ensemble machine learner that constructs multiple decision trees using randomly selected subsets of training data. This approach promotes generalizability and reduces overfitting by ensuring no single tree dominates the learning process. During classification, each tree independently makes predictions, and the model outputs the most common classification among all trees. Random Forest was chosen for its robustness in handling class imbalances and consistent performance in classification tasks.

Logistic Regression [22] is a statistical model tailored for binary classification problems. It models the relationship between features and outcomes using a logarithmic function, making it a simple and interpretable classification technique. Due to its speed, simplicity, and ease of deployment, Logistic Regression is widely used across fields such as finance, social sciences, and healthcare.

V. RESULTS AND ANALYSIS

Our evaluation of data preprocessing methods reveals notable performance differences across machine learning models. XGBoost and LightGBM consistently achieved the highest scores across class balances and categorical data transformation techniques. Their F1 accuracy scores ranged

from 0.79 to 0.81, with AUROC above 0.96 and AUPRC exceeding 0.88 across all conditions. The best-performing model overall was LightGBM at a 75:25 sampling ratio with one-hot encoding, yielding an F1-score of 0.8159, AUROC of 0.9691, and AUPRC of 0.8954.

Random Forest showed slightly lower performance, with F1 accuracy scores ranging from 0.77 to 0.79, AUROC around 0.95, and AUPRC around 0.87. However, performance declined slightly when the 50:50 class balance was reached. Logistic Regression had the lowest performance, with its best F1-scores at 0.72 to 0.73, AUROC between 0.94 and 0.95, and AUPRC near 0.83 in the initial imbalanced dataset. Its performance steadily declined as the class distribution became more balanced.

Adjusting class balance ratios impacted models differently. Boosting algorithms XGBoost and LightGBM exhibited minimal performance change, with only a slight 0.01 decrease across F1, AUROC, and AUPRC scores. This resilience may be attributed to their robust design, which integrates multiple learners for enhanced generalizability [23]. Random Forest maintained F1-scores between 0.77 and 0.79, AUROC between 0.94 and 0.95, and AUPRC between 0.86 and 0.87, except at the 50:50 class balance, where performance dropped to F1-scores between 0.74 and 0.77, with slight AUROC and AUPRC reductions. This decrease may stem from the loss of real training samples that is needed to create the 50:50 class balance.

Logistic Regression showed a consistent decline in F1, AUROC, and AUPRC as class balance increased. This may be attributed to non-ensemble models like Logistic Regression lacking the mechanisms for generalization, making them more sensitive to class distribution shifts.

One-hot encoding and Weights of Evidence (WOE) demonstrated comparable performance across F1, AUROC, and AUPRC metrics, with no clear pattern favoring one technique over the other. Information Value (IV)-based feature elimination had no measurable impact on performance, as models maintained similar predictive strength before and after feature removal. Across all models, training and test metrics remained closely aligned, with only minor decreases from training to test sets, indicating strong generalization and minimal overfitting.

Examining feature importance rankings across different categorical transformation techniques reveals consistent patterns in influential predictors. HBA1C and blood glucose emerge as the two most critical features across all rankings, demonstrating a strong association with diabetes prediction. Age and smoking history also maintain high importance, suggesting their relevance in assessing diabetic risk. The WOE with IV approach eliminates gender and race from its rankings, indicating that IV based feature selection reduces reliance on demographic variables. In contrast, both WOE without IV and one-hot encoding retain Gender and Race, though they rank lower in importance than medical tests.

VI. CONCLUSION AND FUTURE WORK

This comparative analysis highlights the robustness of tree-based algorithms such as gradient boosting methods XGBoost, LightGBM and Random Forest, to perform consistently across varying data conditions. In contrast, Logistic Regression exhibited greater sensitivity to changes

in class balance, suggesting that in the case of complex and imbalanced datasets, such as diabetes related clinical data, tree-based ensemble models may be preferable over simpler machine learning approaches.

The comparable performance of Weights of Evidence (WOE) and one-hot encoding underscores WOE's potential as an effective categorical data transformation method in diabetes analytics. WOE may outperform one-hot encoding in datasets with high-cardinality categorical features, where one-hot encoding can introduce sparsity. Further exploration could provide deeper insights into their comparative performance across different datasets.

Additionally, our feature importance analysis reaffirms the dominance of biochemical markers, such as HBA1C, blood glucose, BMI, and age as influential predictors in diabetes classification across preprocessing techniques. The relevance of smoking history also underscores the importance of lifestyle factors in predictive modeling. Future studies could leverage these insights to refine feature selection strategies in healthcare applications.

While this study focuses on shallow machine learning techniques, future research could examine deep learning approaches to assess how model performance interacts with class balance, categorical data transformation methods, and feature importance in neural networks. These investigations may yield different insights and optimization strategies for predictive modeling in healthcare applications.

TABLE III. WOE AND IV PERFORMANCE METRICS

XGBoost	Acc	Prec	Rec	F1	G-Mean	AUROC	AUPRC
0	1						
85.77%	14.28%	0.9495	0.9379	0.6927	0.7968	0.8291	0.9639
80.36%	19.64%	0.9512	0.9370	0.7058	0.8042	0.8368	0.9652
75%	25%	0.9495	0.9201	0.7047	0.7978	0.8333	0.9637
62.50%	37.50%	0.9489	0.8999	0.7172	0.8001	0.8440	0.9662
50%	50%	0.9439	0.8507	0.7310	0.7873	0.8456	0.9658
LightGBM							
0	1						
85.77%	14.28%	0.9515	0.9698	0.6811	0.8011	0.8242	0.9683
80.36%	19.64%	0.9530	0.9726	0.6977	0.8121	0.8323	0.9681
75%	25%	0.9523	0.9565	0.7012	0.8073	0.8356	0.9673
62.50%	37.50%	0.9501	0.9227	0.7059	0.8009	0.8364	0.9675
50%	50%	0.9481	0.8841	0.7228	0.8004	0.8395	0.9671
Random Forest							
0	1						
85.77%	14.28%	0.9494	0.9341	0.6906	0.7961	0.8299	0.9559
80.36%	19.64%	0.9486	0.9176	0.7013	0.7955	0.8327	0.9545
75%	25%	0.9464	0.9019	0.7053	0.7922	0.8346	0.9546
62.50%	37.50%	0.9447	0.8649	0.7225	0.7896	0.8411	0.9555
50%	50%	0.9382	0.8069	0.7392	0.7738	0.8474	0.9561
Logistic Regression							
0	1						
85.77%	14.28%	0.9317	0.8480	0.6557	0.7286	0.7983	0.9434
80.36%	19.64%	0.9270	0.7707	0.7057	0.7348	0.8240	0.9456
75%	25%	0.9131	0.6850	0.7523	0.7115	0.8368	0.9441
62.50%	37.50%	0.8648	0.5156	0.8348	0.6382	0.8524	0.9439
50%	50%	0.7941	0.4049	0.9112	0.5600	0.8412	0.9423

XGBoost		Acc	Prec	Rec	F1	G-Mean	AUROC	AUPRC
0	1							
85.77%	14.28%	0.9522	0.9480	0.7132	0.8107	0.8395	0.9658	0.8906
80.36%	19.64%	0.9510	0.9373	0.7026	0.8040	0.8360	0.9654	0.8856
75%	25%	0.9500	0.9302	0.7008	0.7996	0.8327	0.9648	0.8833
62.50%	37.50%	0.9477	0.8945	0.7197	0.7965	0.8415	0.9640	0.8839
50%	50%	0.9417	0.8323	0.7389	0.7852	0.8514	0.9643	0.8827
LightGBM								
0	1							
85.77%	14.28%	0.9519	0.9646	0.6892	0.8039	0.8284	0.9671	0.8891
80.36%	19.64%	0.9517	0.9617	0.6924	0.8032	0.8291	0.9670	0.8905
75%	25%	0.9530	0.9607	0.7011	0.8098	0.8349	0.9678	0.8907
62.50%	37.50%	0.9513	0.9298	0.7114	0.8071	0.8407	0.9672	0.8913
50%	50%	0.9462	0.8738	0.7310	0.7974	0.8478	0.9656	0.8878
Random Forest								
0	1							
85.77%	14.28%	0.9460	0.8994	0.6973	0.7873	0.8293	0.9447	0.8565
80.36%	19.64%	0.9447	0.8791	0.7072	0.7826	0.8322	0.9477	0.8606
75%	25%	0.9433	0.8684	0.7147	0.7836	0.8387	0.9487	0.8646
62.50%	37.50%	0.9433	0.8684	0.7147	0.7836	0.8387	0.9487	0.8646
50%	50%	0.9248	0.7491	0.7735	0.7529	0.8590	0.9548	0.8658
Logistic Regression								
0	1							
85.77%	14.28%	0.9340	0.8601	0.6482	0.7386	0.8003	0.9448	0.8321
80.36%	19.64%	0.9267	0.7742	0.6899	0.7304	0.8229	0.9440	0.8206
75%	25%	0.9179	0.6958	0.7562	0.7232	0.8444	0.9476	0.8341
62.50%	37.50%	0.8646	0.5167	0.8312	0.6379	0.8509	0.9422	0.8240
50%	50%	0.7899	0.3958	0.9069	0.5507	0.8361	0.9404	0.8240

XGBoost		Acc	Prec	Rec	F1	G-Mean	AUROC	AUPRC
0	1							
85.77%	14.28%	0.9515	0.9495	0.6980	0.8044	0.8328	0.9651	0.8851
80.36%	19.64%	0.9504	0.9355	0.7012	0.8015	0.8339	0.9660	0.8868
75%	25%	0.9505	0.9321	0.7059	0.8028	0.8341	0.9655	0.8860
62.50%	37.50%	0.9505	0.9321	0.7059	0.8028	0.8341	0.9655	0.8860
50%	50%	0.9443	0.8571	0.7321	0.7897	0.8468	0.9650	0.8839
LightGBM								
0	1							
85.77%	14.28%	0.9523	0.9693	0.6880	0.8061	0.8270	0.9663	0.8892
80.36%	19.64%	0.9526	0.9721	0.6906	0.8063	0.8283	0.9673	0.8929
75%	25%	0.9546	0.9676	0.7049	0.8159	0.8382	0.9691	0.8954
62.50%	37.50%	0.9508	0.9393	0.7027	0.8039	0.8344	0.9664	0.8884
50%	50%	0.9474	0.8892	0.7260	0.7999	0.8496	0.9661	0.8897
Random Forest								
0	1							
85.77%	14.28%	0.9486	0.9282	0.6938	0.7940	0.8292	0.9436	0.8600
80.36%	19.64%	0.9468	0.9096	0.6966	0.7895	0.8303	0.9456	0.8592
75%	25%	0.9454	0.8898	0.7087	0.7882	0.8335	0.9492	0.8638
62.50%	37.50%	0.9392	0.8132	0.7370	0.7730	0.8464	0.9504	0.8636
50%	50%	0.9266	0.7295	0.7559	0.7438	0.8502	0.9521	0.8579
Logistic Regression								
0	1							
85.77%	14.28%	0.9353	0.8729	0.6410	0.7392	0.7941	0.9461	0.8337
80.36%	19.64%	0.9244	0.7544	0.6923	0.7202	0.8141	0.9415	0.8146
75%	25%	0.9157	0.7412	0.6838	0.7063	0.8042	0.9314	0.7939
62.50%	37.50%	0.8971	0.6173	0.7521	0.6819	0.8314	0.9315	0.8076
50%	50%	0.8932	0.5941	0.7735	0.6646	0.8347	0.9328	0.8010

WOE NO IV		WOE WITHIV		ONE HOT	
HBA1C	3.2372	HBA1C	3.1466	HBA1C	3.1971
Blood Glucose	1.9690	Blood Glucose	1.9916	Blood Glucose	1.9961
Age	0.1877	BMI	0.1990	BMI	0.1872
BMI	0.1871	Age	0.1836	Age Bucket: 40-50	0.0949
Smoking History	0.0872	Smoking History	0.0920	Smoking History: No Info	0.0723
Hypertension	0.0709	Hypertension	0.0691	Hypertension	0.0716
Gender	0.0615	Heart Disease	0.0540	Age Bucket: 70-81	0.0567
Heart Disease	0.0475			Heart Disease	0.0523
Race	0.0299			Age Bucket:150-60	0.0466
				Gender: Male	0.0350
				Gender: Female	0.0250
				Age Bucket: 60-70	0.0232
				Race: Other	0.0124
				Race: African American	0.0113
				Smoking History: Never	0.0110
				Smoking History: Ever	0.0107
				Smoking History: Former	0.0101
				Smoking History: Not Current	0.0099
				Race: Asian	0.0088
				Race: Hispanic	0.0060
				Race: Caucasian	0.0050
				Smoking History: Current	0.0044
				Gender: Other	0.0000

WOE NO IV		WOE WITHIV		ONE HOT	
HBA1C	0.5687	HBA1C	0.5747	HBA1C	0.5696
Blood Glucose	0.3326	Blood Glucose	0.3355	Blood Glucose	0.3328
BMI	0.0439	BMI	0.0447	BMI	0.0433
Age	0.0191	Age	0.0195	Hypertension	0.0090
Smoking History	0.0099	Smoking History	0.0104	Heart Disease	0.0088
Hypertension	0.0077	Hypertension	0.0077	Age Bucket: 40-50	0.0063
Heart Disease	0.0073	Heart Disease	0.0075	Age Bucket: 70-81	0.0042
Race	0.0069			Smoking History: No Info	0.0042
Gender	0.0040			Age Bucket: 50-60	0.0037
				Gender: Female	0.0028
				Age Bucket: 60-70	0.0023
				Smoking History: Never	0.0015
				Smoking History: Former	0.0014
				Race: Other	0.0013
				Race: African American	0.0013
				Race: Caucasian	0.0012
				Gender: Male	0.0012
				Race: Hispanic	0.0012
				Smoking History: Not Current	0.0012
				Race: Asian	0.0009
				Smoking History: Current	0.0009
				Smoking History: Ever	0.0008
				Gender: Other	0.0000

WOE NO IV		WOE WITHIV		ONE HOT	
HBA1C	2.9596	HBA1C	2.9632	HBA1C	2.9664
Blood Glucose	2.2358	Blood Glucose	2.1671	Blood Glucose	2.1919
BMI	0.3842	BMI	0.3844	BMI	0.3842
Age	0.2722	Age	0.2720	Age Bucket: 40-50	0.1863
Smoking History	0.2011	Smoking History	0.1920	Age Bucke:t 50-60	0.1308
Gender	0.1205	Hypertension	0.1129	Smoking History: No Info	0.1260
Hypertension	0.1191	Heart Disease	0.1046	Hypertension	0.1083
Race	0.1110			Gender: Male	0.0956
Heart Disease	0.0973			Age Bucket: 70-81	0.0951
				Heart Disease	0.0938
				R ace: African American	0.0535
				Smoking History: Never	0.0480
				Race: Other	0.0448
				Gender: Female	0.0448
				Age Bucket: 60-70	0.0443
				Smoking History: Former	0.0384
				Race: Caucasian	0.0383
				Smoking History: Not Current	0.0369
				Smoking History: Ever	0.0337
				Race: Asian	0.0333
				Race: Hispanic	0.0325
				Smoking History: Current	0.0218
				Gender: Other	0.0000

TABLE IX. XGBOOST BUILT-IN

WOE NO IV		WOE WITH IV		ONE HOT	
HBA1C	0.5392	HBA1C	0.5537	HBA1C	0.4328
Blood Glucose	0.2610	Blood Glucose	0.2700	Blood Glucose	0.2052
Hypertension	0.0487	Hypertension	0.0514	Age Bucket: 40-50	0.0408
Heart Disease	0.0425	Heart Disease	0.0475	Gender: Male	0.0389
Age	0.0350	Age	0.0374	Heart Disease	0.0373
BMI	0.0211	BMI	0.0219	Hypertension	0.0332
Smoking History	0.0198	Smoking History		Smoking History: No Info	0.0328
Gender	0.0191			Age Bucket: 50-60	0.0236
Race	0.0136			Age Bucket: 70-81	0.0178
				BMI	0.0158
				Smoking History: Former	0.0120
				Gender: Female	0.0109
				Race: Hispanic	0.0109
				Smoking History: Ever	0.0109
				Smoking History: Current	0.0107
				Age Bucket: 60-70	0.0101
				Smoking History: Not Current	0.0100
				Race: Other	0.0097
				Smoking History: Never	0.0097
				Race: Asian	0.0094
				Race: African American	0.0091
				Race: Caucasian	0.0084
				Gender: Other	0.0000

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